Michael Rizzo-Lopez

(previously Michael Rizzo Smith)

Curriculum Vitae

Nashville, TN — michael.rizzolopez@vanderbilt.edu — mjosmith.github.io

Computational astrophysicist specializing in hydrodynamic simulations of massive black hole binary dynamics in gaseous accretion disks and predicting their electromagnetic and gravitational-wave signatures for multi-messenger astrophysics.

EDUCATION

Ph.D Astrophysics — Aug. 2023 - Exp. 2029
Vanderbilt University — Nashville, TN
Department of Physics and Astronomy
Dissertation Title: *"Hydrodynamic Simulations of Relativistic Compact Object Inspirals in Accretion Disks"*
Advisor: *Kelly Holley-Bockelmann*

Bachelor of Science, Physics and Astronomy & Planetary Science — Aug. 2017 - May 2021
Stony Brook University — Stony Brook, NY
Department of Physics and Astronomy
Magna Cum Laude with Honors in Physics

PROFESSIONAL WORK AND AFFILIATIONS

LISA Early Career Scientist Working Group, Chair — Aug. 2026 - Present
NASA XRP 2026 Review Panel, Executive Secretary — Mar. 2026
LISA Consortium, Core Member — May 2025 - Present
Vanderbilt's Graduate Student Astrophysics Journal Research Club, Co-Coordinator — Jan. 2025 - Aug. 2026
Rebuilt existing club structure to emphasize practical presentation skills and implemented a formal faculty feedback structure.
LISA Disc-IMRI Project Astrophysics Working Group, Code Captain — Dec. 2023 - Dec. 2025
Establishing Multimessenger Astronomy Inclusive Training (EMIT), Graduate Student Scholar — Sep. 2023 - Present
ASAS-SN Project, Data Analyst — Jun. 2021 - July 2023
Research Assistant for the All Sky Automated Survey for Supernovae at The Ohio State University.

AWARDS AND FELLOWSHIPS

Vanderbilt Physics and Astronomy Summer Research Award (*$2,000*) — May 2026
NASA CRESST II LISA Symposium Travel Award (*$1,500*) — Apr. 2026
NSF Graduate Research Fellow (*$150,000*) — Jun. 2025
Vanderbilt Graduate School Travel Grant (*$1,000*) — July 2024
McMinn Research Award (*$2,000*) — May 2024
Vanderbilt Robert T. Lagemann Award (*$3,500*) — Apr. 2024
Departmental Honors, Dept. of Physics — May 2021
Stony Brook Campus Scholarship (*$26,000*) — Aug. 2017 - May 2021

PRESENTATIONS

Contributed Talk, The 16th International LISA Symposium — June 2026
Presenting: *"Precessing, Eccentric Binary Black Hole Evolution in Accretion Disks"*
Contributed Talk, Vanderbilt's Astrophysics Journal Research Club — Nov. 2025
Presenting: *"Relativistic Binary Precession: Impact on Eccentric Massive Binary Black Hole Accretion and Hydrodynamics"*
Contributed Talk, Vanderbilt's Astrophysics Journal Research Club — Jan. 2025
Presenting: *"The dynamics and electromagnetic signatures of accretion in unequal mass binary black hole inspirals"*
Contributed Talk, The 15th International LISA Symposium — July 2024
Presenting on behalf of the Disc-IMRI Code Comparison LISA Astrophysics Working Group

PUBLICATIONS

Reviewed Articles

(1 first-author, 5 co-author, 141 citations)

1. Andrea Derdzinski, Alexander J. Dittmann, Alessia Franchini, Alessandro Lupi, Noé Brucy, Pedro R. Capelo, Frédéric S. Masset, Raphaël Mignon-Risse, **Michael Rizzo Smith**, Edwin Santiago-Leandro, Martina Toscani, David A. Velasco-Romero, Robert Wissing, Mudit Garg, Lucio Mayer, Roberto Serafinelli, Lazaros Souvaitzis, Daniel J. D'Orazio, and Jonathan Menu. "The LISA Astrophysics "Disc-IMRI" Code Comparison Project: Intermediate-Mass-Ratio Binaries in AGN-Like Discs". In: *arXiv e-prints* (Dec. 2025), arXiv:2512.10893. arXiv: `2512.10893 [astro-ph.HE]`
2. Kelly Holley-Bockelmann, Fazeel Mahmood Khan, Isaiah Williams, Jaelyn Roth, **Rizzo Smith, Michael**, Kaitlin Porter, Jillian Bellovary, Andrea Derdzinski, and Andrea V. Macciò. "Handy Relation between Binary Black Hole Merger Times and Host Galaxy Properties". In: *Astrophysical Journal Letters* 995.1, L32 (Dec. 2025), p. L32. arXiv: `2508.14253 [astro-ph.GA]`
3. J. M. M. Neustadt, C. S. Kochanek, and **Rizzo Smith, M.** "Constraints on pre-SN outbursts from the progenitor of SN 2023ixf using the large binocular telescope". In: *MNRAS* 527.3 (Jan. 2024), pp. 5366–5373. arXiv: `2306.06162 [astro-ph.HE]`
4. Matthew Kenworthy, [20 authors], and **Rizzo Smith, Michael**. "A planetary collision afterglow and transit of the resultant debris cloud". In: *Nature* 622.7982 (Oct. 2023), pp. 251–254. arXiv: `2310.08360 [astro-ph.EP]`
5. **Rizzo Smith, M.**, C. S. Kochanek, and J. M. M. Neustadt. "The late time optical evolution of twelve core-collapse supernovae: detection of normal stellar winds". In: *MNRAS* 523.1 (July 2023), pp. 1474–1495. arXiv: `2212.09763 [astro-ph.HE]`
6. A. Kawash, [11 authors], **Rizzo Smith, M.**, T. W. -S. Holoien, J. L. Prieto, and T. A. Thompson. "The Galactic Nova Rate: Estimates from the ASAS-SN and Gaia Surveys". In: *The Astrophysical Journal* 937.2, 64 (Oct. 2022), p. 64. arXiv: `2206.14132`

Non-Reviewed Articles

(6 first-author, 13 citations)

1. M. Rizzo Smith and ASASSN-Team. "An Update on ASASSN-21qj: A Rapidly Fading, Sun-Like Star; Back With a Vengeance". In: *The Astronomer's Telegram* 15531 (July 2022), p. 1
2. M. Rizzo Smith and ASASSN-Team. "ASASSN-22el: A Deep Eclipse Event". In: *The Astronomer's Telegram* 15308 (Apr. 2022), p. 1
3. M. Rizzo Smith and ASASSN-Team. "ASASSN-21sa: A Deep Dimming Event". In: *The Astronomer's Telegram* 14937 (Sept. 2021), p. 1
4. M. Rizzo Smith and ASASSN-Team. "ASASSN-21qj: A Rapidly Fading, Sun-Like Star". In: *The Astronomer's Telegram* 14879 (Aug. 2021), p. 1
5. M. Rizzo Smith and ASASSN-Team. "ASASSN-21nn: An Unusual Dimming Event in a Red Giant Star". In: *The Astronomer's Telegram* 14803 (July 2021), p. 1
6. M. Rizzo Smith and ASASSN-Team. "ASASSN-21ml: Discovery of an Extreme (Delta g > 10 mag) L-Dwarf Flare". In: *The Astronomer's Telegram* 14778 (July 2021), p. 1

OUTREACH

Nashville Adventure Science Center, Way Late Play Date, Invited Speaker — Apr. 2026
Public fundraising event dedicated to making science learning accessible for all.
"Seeing" Black Holes (50+ attendees)

Astronomy on Tap Nashville, Host — 2025 - Present
Facilitate event, run trivia, and answer astronomy related questions from the public.
(100+ monthly attendees)

10-Years of Gravitational Waves, Vanderbilt Community Celebration, Poster — Sep. 2025
"It's OK to Be Gassy: How Cosmic Gas Guides Black Hole Collisions" (50+ attendees)

NASA Advocacy Day, Vanderbilt Federal Relations — July 2025
Met with congressional staffers in Washington D.C. to emphasize funding public science and NASA.

NSF GRFP: Navigating the Application Panel, Panelist — Oct. 2025
Panel to prospective applicants (30+ attendees).

Vanderbilt Society of Physics Students, Invited Panelist — Oct. 2025
Panel on graduate school application process (15 students).

EMIT Outreach,Invited Speaker — Jan. 2025
Career talk on computational astrophysics and higher ed (50+ students).

Vanderbilt Society of Physics Students, Invited Panelist — Oct. 2024
Panel on graduate school application process (20+ students).

Valor Collegiate Academy,Invited Speaker — Mar. 2024
Career talk to first-year class (150+ students).

TEACHING

Graduate Teaching Assistant Fall 2023 - Spring 2025

Course Title: ASTR 1010L: Intro to Astronomy Lab

Responsibilities: Course instructor for weekly 3.5 hour class, lectured, facilitated group work, and graded lab packets.

Number of Students: ∼21 per term, 4 terms – Rated 5/5 Effective Instructor by 100% of Evaluation Respondents (∼ 50% of students taught).

Private Tutor Feb. 2020 - Oct. 2020

Intro Physics for Medical Professionals

Undergraduate Teaching Assistant Spring 2020

Course Title: AST 203: Astronomy